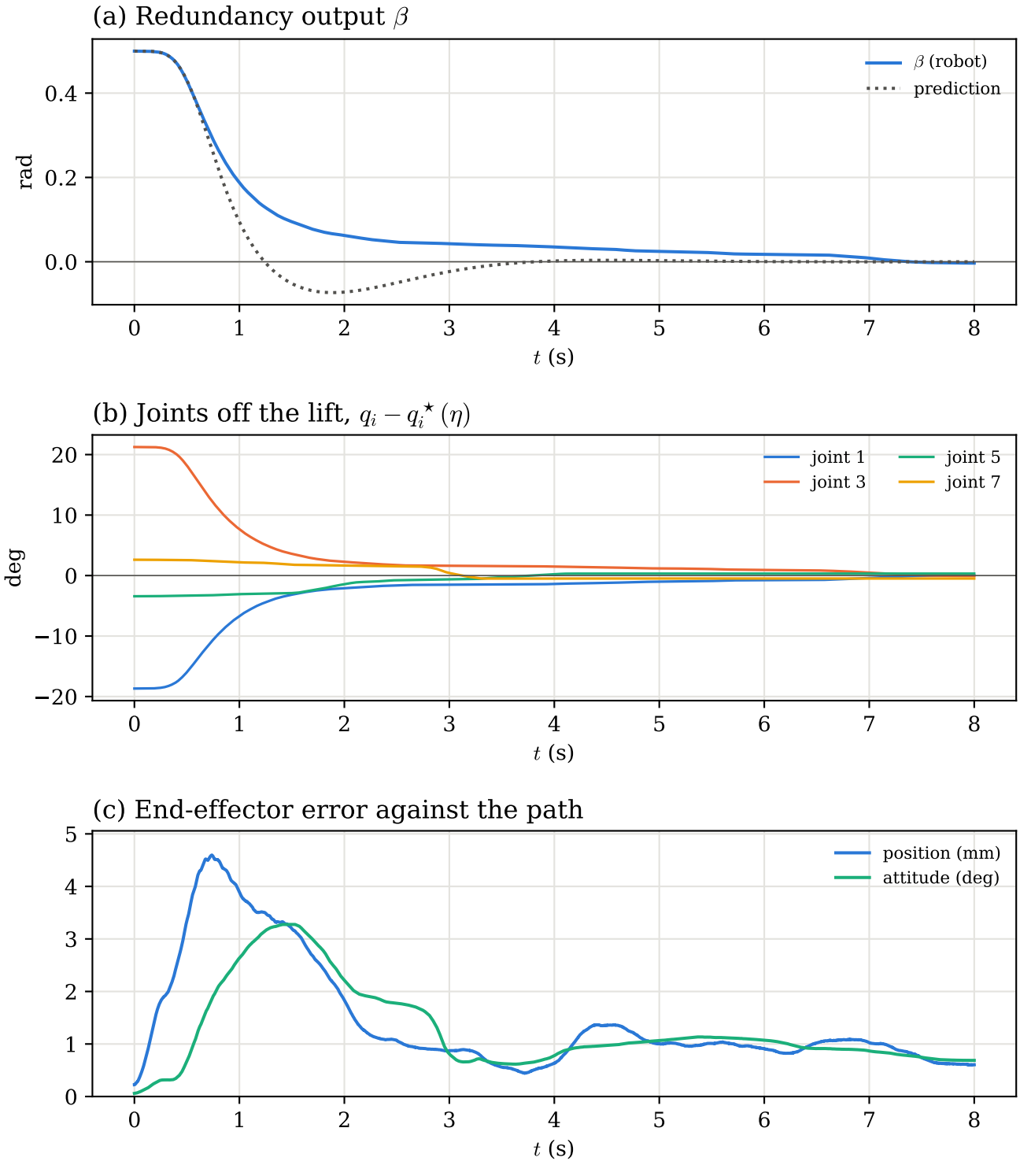
(a) Redundancy output β
β (robot)
prediction
rad
0.4
0.2
0.0
0
1
2
3
4
5
6
7
8
t (s)
(b) Joints off the lift, $q_i - q_i^\star$ (η)
joint 1
joint 3
joint 5
joint 7
deg
20
10
0
−10
−20
0
1
2
3
4
5
6
7
8
t (s)
(c) End-effector error against the path
position (mm)
attitude (deg)
5
4
3
2
1
0
0
1
2
3
4
5
6
7
8
t (s)